

HRV (RMSSD) Signal

VRET pipeline • Writer: Shayan Itami

1. What this signal is, and why we use it

Heart-rate variability (HRV) is the variation in time between consecutive heartbeats. We quantify it with **RMSSD** — the root mean square of successive differences between adjacent R-R intervals (the time between successive R-peaks of the ECG). RMSSD is the standard short-term, time-domain HRV index and chiefly reflects parasympathetic (vagal) influence on the heart.

Physiologically, a relaxed person shows **high** beat-to-beat variability (high RMSSD); under stress or arousal the sympathetic system dominates and variability **drops** (RMSSD falls). So in this pipeline, **RMSSD suppression relative to baseline indicates arousal**. HRV carries weight 0.3 in the composite stress score S_t (EDA 0.5, HRV 0.3, HR 0.2). It is the slowest of the three signals to become available, because measuring variability requires many beats.

■ **Reference.** *Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology (1996). Heart rate variability: standards of measurement, physiological interpretation, and clinical use. Circulation 93(5):1043-1065 / Eur Heart J 17(3):354-381. — Defines RMSSD and the conventions used throughout this document. Shaffer & Ginsberg (2017), An overview of heart rate variability metrics and norms, Front. Public Health 5:258 — RMSSD as a vagal-tone index and typical resting ranges.*

2. The signal path at a glance

Raw ECG → **clean** (nk.ecg_clean: highpass + 50 Hz powerline removal) → **R-peak detection** (nk.ecg_peaks) → **Kubios artifact correction** (Lipponen & Tarvainen, applied inside ecg_peaks) → **R-R intervals** → **Malik 20% gate** (reject non-physiological beats) → **RMSSD**. The same gated-RMSSD routine is used identically for the frozen baseline and for every live update, so the two are directly comparable.

■ **Reference.** *R. P. Lipponen & M. P. Tarvainen (2019). A robust algorithm for heart rate variability time series artefact correction using novel beat classification. J. Med. Eng. Technol. 43(3):173-181. — The 'Kubios' artifact-correction method invoked by nk.ecg_peaks(correct_artifacts=True). Makowski et al. (2021), NeuroKit2: a Python toolbox for neurophysiological signal processing, Behav. Res. Methods 53:1689-1696 — the library providing ecg_clean / ecg_peaks / hrv_time.*

3. Configuration constants

The values governing the HRV computation, quoted from the source:

```
HRV_WINDOW_S      = 60    # moving window for RMSSD (statistical smoothing)
HR_COMPUTE_INTERVAL = 25    # recompute HR/RMSSD every N ticks (~0.5 s)
MIN_PEAKS_HRV      = 10
RMSSD_MIN_MS       = 5
RMSSD_MAX_MS       = 300
RR_MIN_MS          = 300    # 200 BPM ceiling
RR_MAX_MS          = 1500   # 40 BPM floor
RR_MAX_RELATIVE_CHANGE = 0.20 # Malik 20% rule
HRV_WIN_SAMPLES     = HRV_WINDOW_S * SAMPLING_RATE_HZ
```

HRV_WINDOW_S = 60. RMSSD is computed over a 60-second moving window. Short-term RMSSD is conventionally measured over windows of roughly 1-5 minutes; 60 s is the accepted short-term floor, long enough to contain enough beats for a stable estimate (~70-100 beats at typical rates) yet short enough to track changes during an exposure session.

RMSSD_MIN_MS / RMSSD_MAX_MS = 5 / 300. A physiological plausibility band. Genuine resting RMSSD sits roughly in the tens of milliseconds; a value far outside 5-300 ms is not a real measurement but a sign the R-peak detector mis-placed beats. Such values are rejected rather than reported.

RR_MIN_MS / RR_MAX_MS = 300 / 1500. Individual R-R intervals are first range-filtered: 300 ms = 200 BPM, 1500 ms = 40 BPM. An interval outside this band cannot be a normal resting heartbeat and is discarded as a detector error before RMSSD is formed.

RR_MAX_RELATIVE_CHANGE = 0.20. The Malik 20% rule: an R-R interval differing from the previous one by more than 20% is treated as a missed/extra/ectopic beat and excluded. A *relative* threshold is sounder than a fixed millisecond one because a tolerable beat-to-beat change scales with heart rate.

■ **Reference.** Task Force (1996), as above — source of the short-term RMSSD windowing convention, the R-R plausibility range, and the relative beat-rejection rule (commonly the 20% / Malik criterion). The 5-300 ms band follows the resting-RMSSD norms summarised in Shaffer & Ginsberg (2017).

4. The shared RMSSD routine — gated_rmssd_from_peaks()

Both the baseline and the live phase obtain RMSSD through this one function, so they are always measured the same way. It takes the R-peak sample indices (already Kubios-corrected upstream) and returns a gated RMSSD, or None if the data is too poor.

```
def gated_rmssd_from_peaks(peak_indices):
    rr = np.diff(np.asarray(peak_indices)) * 1000.0 / SAMPLING_RATE_HZ
    rr = rr[(rr >= RR_MIN_MS) & (rr <= RR_MAX_MS)]
    if rr.size < 2:
        return None
    rel_change = np.abs(np.diff(rr)) / rr[:-1]
    diffs = np.diff(rr)[rel_change <= RR_MAX_RELATIVE_CHANGE]
    if diffs.size < (MIN_PEAHS_HRV - 1):
        return None
    r = float(np.sqrt(np.mean(diffs ** 2)))
    if not (np.isfinite(r) and RMSSD_MIN_MS <= r <= RMSSD_MAX_MS):
        return None
    return r
```

Line 1 — R-R intervals in milliseconds. np.diff of the R-peak sample indices gives inter-beat gaps in samples; multiplying by 1000/sampling-rate converts to milliseconds. These are the R-R intervals RMSSD is built from.

Line 2 — range filter. Keep only intervals in the 300-1500 ms (40-200 BPM) physiological band; anything outside is a detector error, removed before it can distort RMSSD.

Lines 5-6 — the Malik 20% gate. rel_change is each successive interval's change relative to the previous interval. We keep only those successive differences whose relative change is within 20%; a larger jump between adjacent beats is a missed or spurious beat, not genuine beat-to-beat variability. This is the step that prevents a handful of bad beats from inflating RMSSD.

Lines 7-8 — sufficiency check. Require at least MIN_PEAHS_HRV-1 surviving differences; too few and the estimate is unreliable, so we return None.

Line 9 — the RMSSD formula itself. Root mean square of the surviving successive differences: sqrt(mean(diff²)). This is the textbook definition of RMSSD, applied to the cleaned interval set.

Lines 10-11 — plausibility ceiling. Reject a result outside 5-300 ms or non-finite; otherwise return the value.

Why not simply call nk.hrv_time() on the corrected peaks? NeuroKit's correct_artifacts=True runs the Kubios method, which is a *rate-reconstruction* algorithm: it fills a missed beat by inserting an interpolated midpoint peak. That keeps the heart-rate trace smooth, but the interpolated interval remains in the series and still inflates a time-domain RMSSD (because RMSSD squares successive differences, a few large interpolated steps dominate). On one validation recording, raw hrv_time gave ~78-93 ms where the gated value was ~40 ms. An inflated, frozen baseline would then make every normal live RMSSD look 'suppressed', manufacturing false arousal for the whole session. Kubios is therefore kept for what it is good at (beat correction) and the Malik gate is added specifically for the RMSSD calculation.

■ **Reference.** Malik / Task Force (1996) for the 20% relative-difference rejection rule and the RMSSD definition. Lipponen & Tarvainen (2019) for the Kubios correction whose reconstruction behaviour motivates the additional gate.

5. Baseline computation (the frozen reference)

During the operator-initiated baseline phase the whole ECG is cleaned once, R-peaks are detected with Kubios correction, and a single frozen avg_hrv is produced via the same gated routine:

```
ecg_all = nk.ecg_clean(ecg_all_raw, sampling_rate=SAMPLING_RATE_HZ)
peaks_all, info_all = nk.ecg_peaks(ecg_all, sampling_rate=SAMPLING_RATE_HZ,
                                   correct_artifacts=True)

...
avg_hrv = gated_rmssd_from_peaks(info_all["ECG_R_Peaks"])
if avg_hrv is None:
    raise RuntimeError(... ECG too noisy to establish a reliable HRV baseline ...)
print(f"[RR-GATE] baseline RMSSD (Kubios + Malik 20% gate) = {avg_hrv:.1f} ms "
      f"(raw hrv_time would give "
      f"{nk.hrv_time(peaks_all,...)[\"HRV_RMSSD\"].iloc[0]:.1f} ms).")
```

ecg_clean first. nk.ecg_peaks expects a cleaned signal; feeding raw mains-contaminated ECG yields zero or wrong R-peaks. Cleaning (highpass + 50 Hz powerline removal) is therefore mandatory before detection. This is the canonical NeuroKit order: ecg_clean → ecg_peaks → (here) gated RMSSD.

correct_artifacts=True. Runs the Kubios beat correction on the detected peaks before they are used.

avg_hrv = gated_rmssd_from_peaks(...). The frozen baseline RMSSD — the reference every live value is compared against. If it cannot be computed (too few clean beats / implausible value) the program refuses to continue rather than proceed on a bad baseline.

The [RR-GATE] print. Reports the gated baseline RMSSD alongside what the ungated hrv_time would have given, so a noisy baseline (large gap between the two numbers) is immediately visible to the operator.

A separate baseline sweep then establishes the per-signal mean and spread used for z-scoring. It steps a 60 s window across the baseline and, for each position, records the HRV deviation (inverted, so suppression is positive):

```
r = gated_rmssd_from_peaks(ir["ECG_R_Peaks"]) # same gate, per window
...
_raw_hrv_d.append((avg_hrv - r) / avg_hrv * 100) # HRV % (inverted)
...
hrv_mean, hrv_sigma = float(_raw_hrv_d.mean()), float(_raw_hrv_d.std())
hrv_sigma = max(hrv_sigma, HRV_SIGMA_FLOOR_PCT, SIGMA_FLOOR) # floor 5%
```

The inverted delta (avg_hrv - r)/avg_hrv. Because HRV *falls* with arousal, the deviation is defined as baseline-minus-live so that suppression yields a **positive** number, consistent with HR and EDA where increases are positive. hrv_mean and hrv_sigma capture the natural resting spread of this quantity.

The sigma floor (HRV_SIGMA_FLOOR_PCT = 5%). The z-score later divides by hrv_sigma. If a baseline is unusually flat, the measured sigma can be implausibly small, and dividing by it would turn a trivial change into a huge z-score. The floor refuses to believe resting HRV spread is below ~5%; it only engages on degenerate baselines and prints a [SIGMA-FLOOR] warning so a suspect baseline is flagged.

■ **Reference.** Task Force (1996) for the windowing and RMSSD conventions. The inversion and z-scoring are an engineering choice of this pipeline (documented here), built on those standard measurements; the sigma floor is a guardrail set from resting-HRV norms (Shaffer & Ginsberg, 2017), not a tuned parameter.

6. Live computation and the bootstrap

During the live phase RMSSD is recomputed on the rolling 60 s ECG buffer, through the same gate:

```

if len(ecg_arr) >= HRV_WIN_SAMPLES:
    ecg_hrv_win = nk.ecg_clean(ecg_arr, sampling_rate=SAMPLING_RATE_HZ)
    peaks_hrv, info_hrv = nk.ecg_peaks(ecg_hrv_win,
                                       sampling_rate=SAMPLING_RATE_HZ, correct_artifacts=True)
    if len(info_hrv["ECG_R_Peaks"]) >= MIN_PEAHS_HRV:
        rmssd_ms = gated_rmssd_from_peaks(info_hrv["ECG_R_Peaks"])
    else:
        rmssd_ms = None

```

The 60 s gate ($\text{len} \geq \text{HRV_WIN_SAMPLES}$). RMSSD is only attempted once a full 60 s of ECG has accumulated; before that there are simply not enough beats to measure variability.

Same clean → peaks → gate sequence as baseline. Using the identical routine guarantees the live RMSSD and the frozen baseline RMSSD are produced the same way, so their comparison is meaningful.

The HRV delta, and the bootstrap that covers the first 60 s:

```

if rmssd_ms is not None:
    delta_hrv = (avg_hrv - rmssd_ms) / avg_hrv * 100
else:
    delta_hrv = hrv_mean # z-scores to ~0: no assumed deviation yet

```

The delta $(\text{avg_hrv} - \text{rmssd_ms})/\text{avg_hrv} * 100$. Percentage suppression of live RMSSD below the frozen baseline; positive means HRV has dropped (arousal), negative means HRV is above baseline (calmer than rest).

The bootstrap ($\text{delta_hrv} = \text{hrv_mean}$). For roughly the first 60 s of the live phase the RMSSD window has not yet filled, so no live RMSSD exists. Rather than guess, delta_hrv is held at its baseline mean, which z-scores to about zero — i.e. 'assume HRV has not moved from baseline yet'. The HRV term keeps its full 0.3 weight but contributes nothing until a real measurement arrives. The baseline RMSSD window is deliberately *not* slid across the operator gap, whose unknown duration would contaminate it.

■ **Reference.** *Bootstrapping an unavailable estimate to its baseline mean (neutral contribution) is an engineering decision of this pipeline; the 60 s requirement follows directly from the short-term RMSSD windowing convention of Task Force (1996).*

7. How HRV enters the composite score

This section is common to all three signal documents so each stands alone. Each signal is converted to a z-score against its own frozen baseline mean and spread, then weighted:

```

def zscore_s_t(eda_phasic_v, hr_pct, hrv_pct):
    zE = (eda_phasic_v - eda_mean) / eda_sigma
    zH = (hr_pct - hr_mean) / hr_sigma
    zV = (hrv_pct - hrv_mean) / hrv_sigma
    return WEIGHT_EDA*zE + WEIGHT_HRV*zV + WEIGHT_HR*zH # 0.5 / 0.3 / 0.2

```

The HRV z-score $\text{zV} = (\text{hrv_pct} - \text{hrv_mean}) / \text{hrv_sigma}$. Expresses the current HRV deviation in units of its own resting spread. This is why z-scoring matters: the three signals have very different natural scales (EDA in microsiemens, HR and HRV in percent), and without normalising each to its own spread the signal with the largest raw numbers would dominate the score regardless of the intended 0.5/0.3/0.2 weights. After z-scoring, the weights mean what they say.

The instantaneous score is smoothed over a short rolling buffer, then classified into three bands against thresholds derived from the baseline score distribution:

```

if s_t >= thresh_high: state = "ultra-stressed"; unity_cmd = "decrease"
elif s_t >= thresh_mild: state = "stressed"; unity_cmd = "neutral"
else: state = "calm"; unity_cmd = "increase"

```

Below the lower threshold the patient is calm and the VR balloon ascends (exposure advances); between the thresholds it holds; above the upper threshold the balloon descends (exposure backs off). The thresholds sit at the

mild and high multiples of the baseline score's spread.

■ **Reference.** *The weighted-sum fusion and the specific 0.5/0.3/0.2 weights are a design decision of the lab/pipeline (documented here). The practice of z-scoring heterogeneous physiological channels to a common scale before fusion is standard in affective-computing / multimodal arousal work.*

8. Known limitations and failure modes

- **Baseline-quality dependence.** The whole HRV term hinges on a trustworthy frozen baseline. The Malik gate removes a few bad beats from an otherwise clean baseline, but it cannot rescue a *globally* noisy baseline (e.g. poor electrode contact throughout): the inflated reference then makes normal live HRV read as constant suppression. A low ECG-quality baseline should be re-recorded.
 - **The first ~60 s is blind.** During the bootstrap the HRV term contributes nothing; genuine HRV changes in the opening minute are not captured by this channel (HR and EDA still respond).
 - **Band placement is not yet validated.** The pipeline correctly separates aroused from calm recordings, but whether a given percentage of 'ultra-stressed' corresponds to the right absolute arousal level needs a recording with a known, graded arousal protocol to calibrate the thresholds.
 - **RMSSD is parasympathetic-weighted.** RMSSD reflects mainly vagal activity; it is a strong but partial proxy for arousal, which is why it is fused with HR and EDA rather than used alone.
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9. References

- [1] Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology (1996). Heart rate variability: standards of measurement, physiological interpretation, and clinical use. *Circulation* 93(5):1043-1065 (also *Eur Heart J* 17(3):354-381).
- [2] Lipponen, J. A., & Tarvainen, M. P. (2019). A robust algorithm for heart rate variability time series artefact correction using novel beat classification. *Journal of Medical Engineering & Technology* 43(3):173-181.
- [3] Makowski, D., Pham, T., Lau, Z. J., et al. (2021). NeuroKit2: A Python toolbox for neurophysiological signal processing. *Behavior Research Methods* 53:1689-1696.
- [4] Shaffer, F., & Ginsberg, J. P. (2017). An overview of heart rate variability metrics and norms. *Frontiers in Public Health* 5:258.